

PHYS505 Recitation — Feb 24, 2026

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Problem 3.1

Two concentric spheres have radii a, b ($b > a$) and each is divided into two hemispheres by the same horizontal plane. The upper hemisphere of the inner sphere and the lower hemisphere of the outer sphere are maintained at potential V . The other hemispheres are at zero potential. Determine the potential in the region $a \leq r \leq b$ as a series in Legendre polynomials. Include terms at least up to $l = 4$. Check your solution against known results in the limiting cases $b \rightarrow \infty$ and $a \rightarrow 0$.

If we place the origin at the center of the spheres and choose $z = 0$ as the dividing plane, the general solution includes both +ve and -ve powers of r in the radial part since the region of interest is a finite interval that doesn't include the origin

$$\Phi(r, \theta) = \sum_{l=0}^{\infty} \left[A_l r^l + \frac{B_l}{r^{l+1}} \right] P_l(\cos \theta)$$

with the boundary conditions

$$\begin{aligned} \Phi(a, \theta) &= \sum_{l=0}^{\infty} \left[A_l a^l + \frac{B_l}{a^{l+1}} \right] P_l(\cos \theta) = V \left[1 - \Theta \left(\theta - \frac{\pi}{2} \right) \right] \\ \Phi(b, \theta) &= \sum_{l=0}^{\infty} \left[A_l b^l + \frac{B_l}{b^{l+1}} \right] P_l(\cos \theta) = V \Theta \left(\theta - \frac{\pi}{2} \right) \end{aligned}$$

where Θ is the (Heaviside) step function. As usual, we exploit the orthogonality of the Legendre polynomials

$$\int_{-1}^1 P_l(\cos \theta) P_{l'}(\cos \theta) d(\cos \theta) = \frac{2}{2l+1} \delta_{ll'}$$

to isolate the individual coefficients

$$\begin{aligned}\langle P_l | \Phi(a, \theta) \rangle &= \frac{2}{2l+1} \left(A_l a^l + \frac{B_l}{a^{l+1}} \right) = V \int_0^1 P_l(x) dx \\ \langle P_l | \Phi(b, \theta) \rangle &= \frac{2}{2l+1} \left(A_l b^l + \frac{B_l}{b^{l+1}} \right) = V \int_{-1}^0 P_l(x) dx\end{aligned}$$

Note that

$$\int_{-1}^1 P_l(x) dx = 0 = \int_0^1 P_l(x) dx + \int_0^1 P_l(-x) dx, \quad l \neq 0$$

For even $l \neq 0$,

$$P_l(-x) = P_l(x) \rightarrow \int_{-1}^0 P_l(x) dx = \int_0^1 P_l(x) dx = 0$$

From the derivative recursion formula

$$\begin{aligned}\frac{d}{dx} [P_{l+1}(x) - P_{l-1}(x)] &= (2l+1)P_l(x) \\ [P_{l+1}(x) - P_{l-1}(x)]_0^1 &= (2l+1) \int_0^1 P_l(x) dx\end{aligned}$$

and the values of Legendre polynomials at 0 and ± 1

$$P_l(1) = 1, \quad P_l(-1) = (-1)^l, \quad P_{2k+1}(0) = 0, \quad P_{2k}(0) = \frac{(-1)^k (2k)!}{2^{2k} (k!)^2}$$

we have, for odd l

$$\begin{aligned}(4k+3) \int_0^1 P_{2k+1}(x) dx &= P_{2k}(0) - P_{2k+2}(0) \\ &= \frac{(-1)^k (2k)!}{2^{2k} (k!)^2} - \frac{(-1)^{k+1} (2k+2)!}{2^{2k+2} (k+1)!^2} \\ &= \frac{(-1)^k (2k)!}{2^{2k} k!^2} \left[1 + \frac{(2k+1)(2k+2)}{4(k+1)^2} \right] \\ &= \frac{(-1)^k}{2^{2k+1}} \frac{(2k)!}{k!(k+1)!} (4k+3) \\ p_k \equiv \int_0^1 P_{2k+1}(x) dx &= \frac{(-1)^k}{2^{2k+1}} \frac{(2k)!}{k!(k+1)!} = - \int_{-1}^0 P_{2k+1}(x) dx\end{aligned}$$

We can also evaluate the first few integrals to confirm

$$\begin{aligned}
P_0(x) = 1 & \quad \int_0^1 P_0(x) dx = x|_0^1 = 1 \\
P_1(x) = x & \quad \int_0^1 P_1(x) dx = \frac{x^2}{2} \Big|_0^1 = \frac{1}{2} \\
P_3(x) = \frac{1}{2} (5x^3 - 3x) & \quad \int_0^1 P_3(x) dx = \frac{1}{2} \left(5\frac{x^4}{4} - 3\frac{x^2}{2} \right) \Big|_0^1 = -\frac{1}{8} \\
P_5(x) = \frac{1}{8} (63x^5 - 70x^3 + 15x) & \quad \int_0^1 P_5(x) dx = \frac{1}{8} \left(63\frac{x^6}{6} - 70\frac{x^4}{4} + 15\frac{x^2}{2} \right) \Big|_0^1 = \frac{1}{16}
\end{aligned}$$

We can now plug these into the coefficient equations. For $l = 0$,

$$\begin{aligned}
2 \left(A_0 + \frac{B_0}{a} \right) &= V & \rightarrow B_0 = 0 \\
2 \left(A_0 + \frac{B_0}{b} \right) &= V & A_0 = \frac{V}{2}
\end{aligned}$$

for even $l \neq 0$,

$$\begin{aligned}
\frac{2}{2l+1} \left(A_l a^l + \frac{B_l}{a^{l+1}} \right) &= 0 & \rightarrow B_l = 0 \\
\frac{2}{2l+1} \left(A_l b^l + \frac{B_l}{b^{l+1}} \right) &= 0 & A_l = 0
\end{aligned}$$

for odd $l = 2k + 1$,

$$\begin{aligned}
\frac{2}{2l+1} \left(A_l a^l + \frac{B_l}{a^{l+1}} \right) &= p_k V & B_l &= a^{l+1} \left(\frac{2l+1}{2} p_k V - a^l A_l \right) \\
\frac{2}{2l+1} \left(A_l b^l + \frac{B_l}{b^{l+1}} \right) &= -p_k V & B_l &= b^{l+1} \left(-\frac{2l+1}{2} p_k V - b^l A_l \right) \\
\rightarrow A_l &= -\frac{(2l+1)p_k}{2} \frac{b^{l+1} + a^{l+1}}{b^{2l+1} - a^{2l+1}} V & B_l &= \frac{(2l+1)p_k}{2} \frac{b^{l+1}a^{l+1} (b^l + a^l)}{b^{2l+1} - a^{2l+1}} V \\
\frac{A_l}{V} &= (-1)^{k+1} \frac{(2l+1)}{2^{l+1}} \frac{(l-1)!}{k!(k+1)!} \frac{b^{l+1} + a^{l+1}}{b^{2l+1} - a^{2l+1}} & \frac{B_l}{V} &= (-1)^k \frac{(2l+1)}{2^{l+1}} \frac{(l-1)!}{k!(k+1)!} \frac{b^{l+1}a^{l+1} (b^l + a^l)}{b^{2l+1} - a^{2l+1}}
\end{aligned}$$

Writing out the first few terms, with $\alpha = a/b$,

$$\begin{aligned}\frac{\Phi(r, \theta)}{V} &= \frac{1}{2} - \frac{3}{4(1-\alpha^3)} \left[(1+\alpha^2) \frac{r}{b} - (1+\alpha) \left(\frac{a}{r} \right)^2 \right] \cos \theta \\ &+ \frac{7}{16(1-\alpha^7)} \left[(1+\alpha^4) \left(\frac{r}{b} \right)^3 - (1+\alpha^3) \left(\frac{a}{r} \right)^4 \right] \frac{(5 \cos^3 \theta - 3 \cos \theta)}{2} \\ &- \frac{11}{32(1-\alpha^{11})} \left[(1+\alpha^6) \left(\frac{r}{b} \right)^5 - (1+\alpha^5) \left(\frac{a}{r} \right)^6 \right] \frac{(63 \cos^5 \theta - 70 \cos^3 \theta + 15 \cos \theta)}{8} \\ &+ \frac{75}{256(1-\alpha^{15})} \left[(1+\alpha^8) \left(\frac{r}{b} \right)^7 - (1+\alpha^7) \left(\frac{a}{r} \right)^8 \right] P_7(\cos \theta) + \dots\end{aligned}$$

In the limit $b \rightarrow \infty$, $A_l, \alpha \rightarrow 0$ and from the BC at $r = a$, we are left with

$$\begin{aligned}B_l &\rightarrow (-1)^k \frac{(2l+1)}{2^{l+1}} \frac{(l-1)!}{k!(k+1)!} a^{l+1} V \\ \frac{\Phi(r, \theta)}{V} &= \frac{1}{2} + \frac{3}{4} \left(\frac{a}{r} \right)^2 \cos \theta - \frac{7}{16} \left(\frac{a}{r} \right)^4 \frac{(5 \cos^3 \theta - 3 \cos \theta)}{2} + \frac{11}{32} \left(\frac{a}{r} \right)^6 P_5(\cos \theta) + \dots\end{aligned}$$

In the limit $a \rightarrow 0$, $B_l, \alpha \rightarrow 0$ and from the BC at $r = b$, we have

$$\begin{aligned}A_l &\rightarrow (-1)^{k+1} \frac{(2l+1)}{2^{l+1}} \frac{(l-1)!}{k!(k+1)!} \frac{V}{b^l} \\ \frac{\Phi(r, \theta)}{V} &= \frac{1}{2} - \frac{3}{4} \frac{r}{b} \cos \theta + \frac{7}{16} \left(\frac{r}{b} \right)^3 \frac{(5 \cos^3 \theta - 3 \cos \theta)}{2} - \frac{11}{32} \left(\frac{r}{b} \right)^5 P_5(\cos \theta) + \dots\end{aligned}$$

The results in the textbook and lecture notes are given for a sphere of radius a with

$$\begin{aligned}\Phi(a, \theta) &= \begin{cases} V, & 0 \leq \theta < \pi/2 \\ -V, & \pi/2 < \theta \leq \pi \end{cases} = V [1 - 2\Theta(\theta)] \\ \Phi_{in}(r < a) &= V \frac{r}{a} \left[\frac{3}{2} P_1(\cos \theta) - \frac{7}{8} \left(\frac{r}{a} \right)^2 P_3(\cos \theta) + \frac{11}{16} \left(\frac{r}{a} \right)^4 P_5(\cos \theta) + \dots \right] \\ \Phi_{out}(r > a) &= V \left(\frac{a}{r} \right)^2 \left[\frac{3}{2} P_1(\cos \theta) - \frac{7}{8} \left(\frac{r}{a} \right)^2 P_3(\cos \theta) + \frac{11}{16} \left(\frac{r}{a} \right)^4 P_5(\cos \theta) + \dots \right]\end{aligned}$$

In order to convert the inside solution Φ_{in} to the $a \rightarrow 0$ limit in this problem, we first substitute b for the radius and change the sign of the constant,

$$\Phi_{in}(r < b) = -V \frac{r}{b} \left[\frac{3}{2} P_1(\cos \theta) - \frac{7}{8} \left(\frac{r}{b} \right)^2 P_3(\cos \theta) + \frac{11}{16} \left(\frac{r}{b} \right)^4 P_5(\cos \theta) + \dots \right]$$

This expression takes the value $-/+V$ on the upper/lower hemisphere. To bring it up to 0 on the upper hemisphere, we add a constant term

$$\Phi_0(r < b) = V - V \frac{r}{b} \left[\frac{3}{2} P_1(\cos \theta) - \frac{7}{8} \left(\frac{r}{b} \right)^2 P_3(\cos \theta) + \frac{11}{16} \left(\frac{r}{b} \right)^4 P_5(\cos \theta) + \dots \right]$$

which makes it $2V$ on the lower hemisphere. Defining $V_0 = 2V$, the solution takes the same form as the one we found

$$\frac{\Phi_0(r < b)}{2V_0} = 1 - \frac{r}{b} \left[\frac{3}{2} P_1(\cos \theta) - \frac{7}{8} \left(\frac{r}{b} \right)^2 P_3(\cos \theta) + \frac{11}{16} \left(\frac{r}{b} \right)^4 P_5(\cos \theta) + \dots \right]$$

To compare the outside solution to the $b \rightarrow \infty$ limit in this problem, we note that the radius is already a , the sign in the boundary condition is the same, but we still need the constant term and the factor of 2

$$\frac{\Phi_0(r > a)}{2V_0} = 1 + \left(\frac{a}{r} \right)^2 \left[\frac{3}{2} P_1(\cos \theta) - \frac{7}{8} \left(\frac{a}{r} \right)^2 P_3(\cos \theta) + \frac{11}{16} \left(\frac{a}{r} \right)^4 P_5(\cos \theta) + \dots \right]$$